

Bitcoin Price Prediction: Report 1- Data Collection

1 Introduction & Task Selection

This report highlights the data collection and pre-processing process for a daily prediction of Bitcoin Close Prices. This report highlights the challenges of data collection, cleaning and feature extraction. This project will attempt to predict future daily close prices for the cryptocurrency Bitcoin (BTC). Time permitting, this project may be extended to include the task of price direction classification, though only price prediction preparation is covered in this report.

2 Dataset Collection

Various approaches were undertaken in an effort to collect high quality data of a suitable data range for both Bitcoin and supplementary data.

2.1 Bitcoin Data

Many sources offer limited access to Bitcoin Data, with many being restricted to 'free-tier' limits of amount and time. Alpha Vantage was explored as it provides a reliable source for Bitcoin Data. However, Alpha Vantage only provides access to the last 350 days approx. Going forward, this data will be considered, perhaps as a future validation set. A Kaggle dataset [1], provided access to a comprehensive Bitcoin Historical Dataset, offering OHLC (Open, High, Low, Close) Prices at 1 minute intervals and was chosen for this experiment.

2.2 Supplementary Media Data

2.2.1 Reddit

Seroyizhko et. al. discuss the increasing interest in Reddit as a data mining resource since the events of the GameStop Stock [2] and therefore, initially Reddit was thoroughly explored for this project. However, a number of issues are present, preventing the usage of Reddit in this project. As it was extensively researched, the issues are briefly outlined below. Firstly, the Reddit API 'PRAW' allows access to top posts in subreddits, for e.g. 'r/Bitcoin'. However, this was limited to current posts and with no way of accessing any historical posts. One idea would be to start collecting the top Bitcoin posts for each day now for future use. Alternatives researched were 'Pushshift' [3] and 'Google BigQuery' [4] datasets, but all Bitcoin data appears to either be removed or privatized. Datasets, such as the one created by Seroyizhko et. al, that had utilized earlier available methods were investigated, but all of this data was of very short duration (some as little as two

days) and some also of a very low 'Usability' score on Kaggle. Therefore, Reddit data was deemed unsuitable for this current project.

2.2.2 Market News & Google Trends

Alpha Vantage offer a tool to call Market News for certain topics along with a sentiment score [5]. This was initially looked into, and while start and end dates were possible to define, the limit returned a maximum of 1000 results. This meant the resulting data was of as short a time period as two days. No way of modifying the tool was found and so building a dataset that covered at least six months would take huge resources and would have to be collected over a period of time due to Alpha Vantage's free API limits. This type of media, unlike instant social media posts, may reflect slightly old sentiment due to the nature of publishing. While this tool is worth further investigation, it was not used in this project.

Google Trends was also investigated but as they provided monthly data, this data was not included [6]. However, this interface highlighted some of the popular similar searches to Bitcoin, such as Tesla and Nvidia and provided insights for possible macroeconomic indicators.

2.2.3 Twitter Data

A number of datasets and methods were investigated. Similar to PRAW, from research the free tier of the Twitter/X API only gives access to current tweets and therefore is of limited use to this project [7]. However, it could be considered to start collecting Tweets from this API now and storing them for a possible validation set in later stages of this project. Kaggle was turned to as an alternative. Five datasets were investigated and four were rejected (see Appendix). Some of the rejected datasets did cover a large period but were, however, too old. It would be interesting to perform sentiment analysis on these tweets and use this historical data to predict on other historical Bitcoin Data to investigate the predictive power of Tweets. The dataset chosen, 'Bitcoin Tweets' [8], covers a range from 2021-2023 and will be discussed further in Section 3.1.2.

2.2.4 Financial (Alpha Vantage) Data

Due to the limited nature of the social media supplementary data, financial market data was sourced as an alternative. Unlike for Bitcoin prices on Alpha Vantage, most stock prices are available from the creation of the company to the present [2]. This data was used as

it provides a reliable and clean source of data. Work by Erfanian et. al. [9] indicates the benefits of including macroeconomic indicators for Bitcoin prediction, especially market capitalization, the S&P 500, Dow Jones Industrial Average, and Nasdaq Composite. Based on this approach, relevant stock prices and ETFs were obtained from Alpha Vantage's 'listing status.csv', which provides the ticker symbols for data it covers. The following Stocks and ETFs were chosen to be further investigated: *Apple Inc. Stock*, *Nvidia Corp. Stock*, *Fidelity NASDAQ Composite Index ETF*, *Tesla Inc. Stock*, *Coinbase Global Inc. - Class A Stock* and *Paypal Holdings Inc. Stock*. As ETFS, such as the ones included above, follow a broad index of stock, especially technology companies such as Apple, it is likely that multicollinearity will be present in this dataset. This will be further investigate and only the most relevant predictors will be obtained or data subsets will be used for the modeling phase. Additionally, Federal Funds Effective Rate data [10] was also considered. However, this data is only available on a monthly basis and so was excluded for this experiment.

2.3 Ethical Considerations for Data Collection

The use of APIs is preferable over web scraping due to the ethical and regulatory considerations, such as the need to notify users. It is however, always important to consider the ethical implications of using social media content for sentiment analysis and Bitcoin price prediction due to whether such usage falls under 'legitimate purposes' as outlined in GDPR Article 5 [11]. In line with the recommendations for responsible data reuse, as outlined by Rogers et. al [12], any personal data will be anonymised in the cleaning phase and that the re-usage of Kaggle data is within the original intentions of the dataset curators. The recommended data reuse checklist is included in the appendix and is available for any persons who wish to reuse any potential data created in this project.

3 Data Cleaning, Processing & Feature Extraction

3.1 Missing Values & Time Normalization

3.1.1 Bitcoin Data

This dataset provides extensive coverage and a two year period (2022-2024) was chosen for initial exploration. Firstly the Unix Timestamp was converted to UTC (which was noted by the data creator as the time zone). The Alpha Vantage data for stock does not mention the time zone the data was recorded in. However, as it is a NASDAQ-licensed data provider, it is safe to assume that it is US/New York time and that the daily close was recorded at 4pm local time as this is the end of normal trading hours. Therefore to align the data as accurately as possible, the minute data was grouped by day and three close values were obtained: *Close Price at 4pm New York Time*, *Close Price at UTC Mid-*

night and Average Close Price (per day). This means all three versions of the close price can be used in the predictions and are included below in Figure 1.

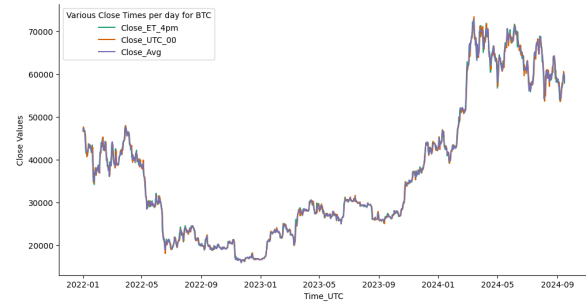


Figure 1: BTC Close Price

It was noted that a number of dates (17/09/2024 - 31/12/2024) appeared to be missing a number of Close values and it is unlikely that all of these days (53) were non-trading days. The data provider mentions that any missing data is due to API or other technical issue, therefore we can assume this data is Missing Completely at Random (MCAR). Due to the sensitivity of the close price fluctuation, it was decided to drop this period and look at the time period from Jan 2022- Aug 2024, as this leaves ample time to investigate long-term seasonal trends in the data.

3.1.2 Twitter Data

As mentioned the dataset was sourced from Kaggle [8]. After converting the dates to US/Eastern time (to align with the Alpha Vantage Data), and sorting by time, we can see the tweets start in early 2021 (05/02/2021) and end in 2023 (09/01/2023). This range should cover 704 days, but only 224 entries are present, meaning 480 dates are missing, which is over two third of the dates. A smaller subset of the data was therefore chosen, with the aim of having a smaller dataset with less missing values. The subset 11/01/2022 - 29/06/2022 was chosen and was found to have 76 entries out of the 170 (45%). While not ideal, it is a vast improvement from earlier. Therefore it was decided to create two datasets, Dataset 1, which represents the Bitcoin and Stock Data from Jan 2022 to mid Sep 2024 and Dataset 2, which represents Bitcoin, Stock and Twitter Data from 11th Jan 2022 to 29th June 2022. Since methods for accessing historical Twitter data is practically non existent, forward fill was used as a placeholder for sentiment for any missing dates.

3.1.3 Financial (Alpha Vantage) Data

Alpha Vantage offer a daily close price for all Stocks/ETFs listed in Section 2.2.4. It was observed that all seven stocks were missing the same number of days. From investigation of January, the days missing were weekend and holiday days. As these dates are Missing at Random (MAR), due to the day of the week, forward fill was chosen to impute these dates. This was

chosen as since the missing days were not trading days, forward filling provides the most likely estimate of the Close price. As the 1st and 2nd of January were missing, the last available date (31/12/2021) had to be extracted from Alpha Vantage and was manually added to the dataset. This successfully filled in any missing Close prices. It should be noted, that this method could potentially introduce bias, especially if the Close prices varied outside the normal trading hours.

3.2 Data Scaling & Feature Extraction

3.2.1 Twitter Data

After time alignment and data subset extraction, the datasets numerous columns were investigated and it was decided to drop the following columns, either for ethical reasons or due to lack of useful information: *user_name* (ethical removal, i.e. anonymisation), *user_location* (either NaN or not reliable), *user_description*, *user_created*, *user_friends*, *user_verified*, *source* (e.g. Web/App), *is_retweet* (all False or NaNs), *user_followers*, *user_favourites*, *hashtags*. While columns such as 'user_followers' may be useful to detect the importance of a certain tweet, since the daily sentiment results will be aggregated and averaged, it was deemed sufficient to keep just the date and text for the purposes of this project. It was also noted that some of the information dropped may be useful for bot detection, however this task is out of the scope of this project. The following steps were taken to preprocess and extract a sentiment score for each tweet:

- **Regex:** Replaces any long URLs and @User mentions for anonymity.
- **NLTK TweetTokenizer:** Tokenizer designed to tokenize tweets specifically (preserves character based emojis) [13].
- **Emoji package:** Converts Emojis to word representations, useful to keep for sentiment analysis [14].
- **VADER:** Extracts Sentiment Intensity and assigns compound score (ranging from -1 to 1) [15].
- **Average:** Daily Average Sentiment Score was calculated.

Other alternatives for sentiment analysis, such as TextBlob and techniques were explored. Future projects could compare the accuracy of various sentiment extraction approaches. However, gold label data would be needed and due to the small dataset present, one technique was deemed sufficient for this project. VADER was chosen due to its training on user generated content, making it suitable for tweet sentiment [15]. The work for this section can be found in the notebook 'Tweet_process_final.ipynb'. While this can be added to the final notebook, it requires high RAM usage and it is recommended to leave it as a separate notebook.

3.2.2 Bitcoin Data & Stock/ETF Data

As the Bitcoin dataset was sourced from an open source medium, it was tested for outliers using box plots and standard deviations. It can be seen in Figure 2 below, that while the data tends to be moving away from the mean, no outliers were detected. This was checked for the average and UTC Close prices.

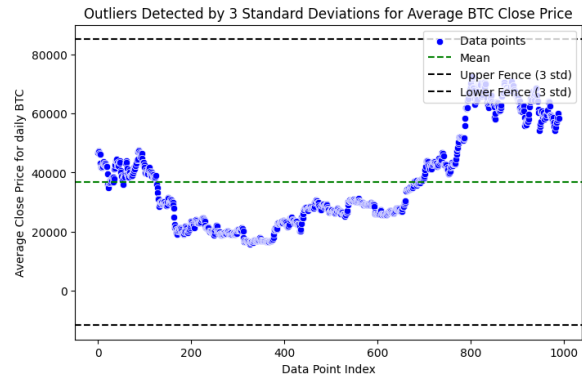


Figure 2: Outlier Detection for BTC Close Price

Double Exponential smoothing (Holt's Method) was applied to the average close price as it can be useful to spot seasonal patterns. As seen below in Figure 3, no immediate pattern can be discerned.

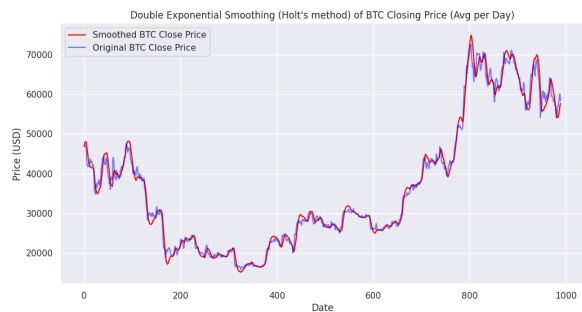


Figure 3: Double Exponential Smoothing for BTC

The Relative Strength Index (RSI) was calculated for Bitcoin close prices as shown in Figure 4 below. It was chosen as it can be a useful tool for investors deciding when to trade by highlighting the strengths and weaknesses of the data [17]. The close price at midnight UTC was chosen for the RSI as it is likely this time will be the future predicted time. This is due to the fact that Alpha Vantage offer BTC Close prices at UTC midnight and therefore can be used to compare the future prediction model.

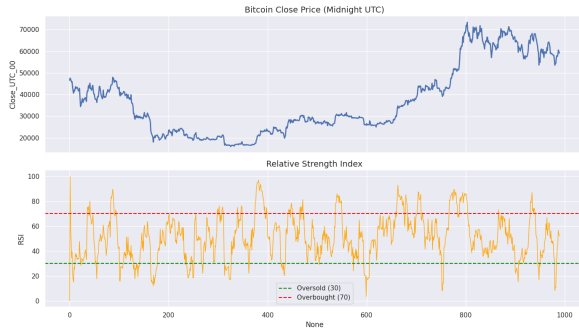


Figure 4: RSI for BTC Close Price

The difference in magnitude between the Bitcoin and Stock Data was noted as well as the difference between both of these and the normalized sentiment score. Recommended by Nair et. al [18], the Min Max Scaler was applied to all data except the sentiment score. This allows for better visualization of all combined trends over time.

4 Exploratory Data Analysis

4.1 Relationships

To explore the effect of all of the various variables, two correlation matrices were created to gain initial insights. From this, it could be seen that RSI, Tesla, Paypal and Nvidia seemed unlikely useful factors for predicting BTC Close prices. The highly correlated features were then extracted and compared against all three close price options to detect any variance. The results are included in table below: It can be seen that

Potential Predictor	Close AVG	Close UTC	Close ET
Sentiment Score	0.63	0.62	0.62
Coinbase	0.88	0.87	0.88
NASDAQ Composite ETF	0.89	0.88	0.90
S&P 500 ETF	0.91	0.90	0.91
Apple	0.92	0.92	0.93

Table 1: Correlation Values for Various Close Price Times

there are slight variations between the various Close Price Columns (UTC vs. ET vs. Avg), and while these changes are small, all will be kept for the modeling phase. Since the variations were small, Close Avg will be used for further relationship investigation.

As mentioned in Section 2.2.4, it is expected that there will be multicollinearity among the potential predictors due to their nature. Therefore, Variance Inflation Factor was used to investigate. This was found to be the case, with all potential predictors exhibiting a VIF greater than 10, with NASDAQ Composite as high as 332. It seems likely that S&P 500 include many of the same stocks as NASDAQ Composite and has a greater correlation to the BTC Close prices. Therefore NASDAQ Composite was dropped and the VIF was recalculated as shown in table 2.

Feature	VIF
const	2041.037717
avg_sentiment_score	2.006479
Close Value Apple	29.517434
Close Value Nvidia	14.421532
Close Value SPDR S&P 500 ETF Trust	35.501754
Close Value Tesla	5.341718
Close Value Coinbase	19.955625
Close Value Paypal	6.129818

Table 2: Variance Inflation Factor (VIF) for Features

It can be seen that this dramatically lowers the VIF, though it is still too high. Nvidia and Paypal were dropped due to their limited correlation to Close Price for Bitcoin. The highly correlated predictors to Bitcoin Close Prices were graphed for better visualization using a correlation matrix as shown in Figure 5 below:

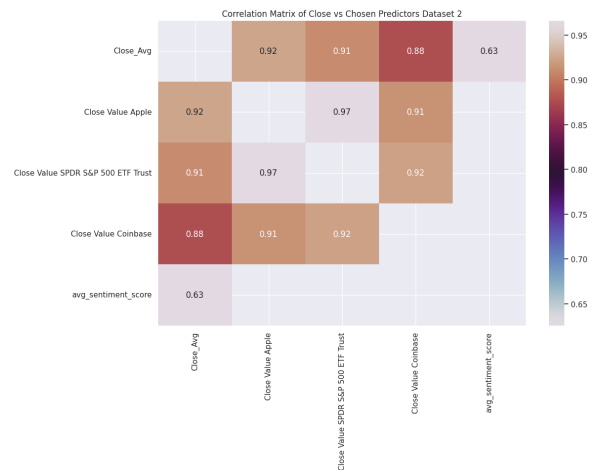


Figure 5: Dataset 2 Potential Predictors

The Corresponding correlation matrix for Dataset 1 is included in the notebook. Due to their high correlation to Close values for BTC, the following features were chosen for the modeling phase: *Sentiment Score*, *Apple Close Values*, *Coinbase Close Values* and *SPDR S&P 500 ETF Values*. These features could all be useful for predicting the Close price of Bitcoin, however, multicollinearity still persists among S&P, Apple and Coinbase. Dropping S&P 500 reduced the VIF to 6 for Apple and Coinbase, which is an acceptable number for modeling. Therefore, for the modeling phase, either separate models will be run in tandem to compare the S&P 500 ETF with the combination of Apple and Coinbase or dimensionality reduction (PCA) could also be used. Finally, some visualization of the trends of the chosen predictors are included below.

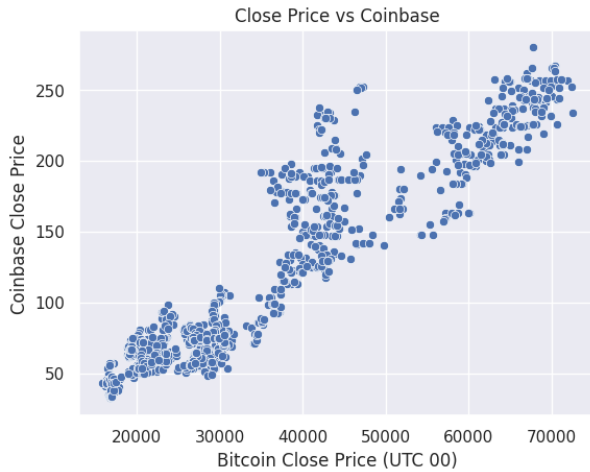


Figure 6: BTC Close vs Coinbase



Figure 7: BTC Close vs Apple

Visualizations of all predictors are included in the final notebook. Note, both notebooks require Google Drive mounting and are designed to create and save the datasets as CSV files to the desired location. This and the inclusion of an Alpha Vantage API are required to replicate the notebook. As mentioned the notebook, 'Tweet_process_final.ipynb' requires extensive RAM usage but it can be run both on a simple GPU or GPU. It is currently set to connect to a T4 GPU. Data from this notebook is called 'tweet_daily_sentiment_final.csv' and is loaded into the final notebook 'Data_Final.csv'. The final datasets created are called 'scaled_data_1_final.csv' and 'scaled_data_2_final.csv' respectively. All notebooks and datasets can be found in the zipped folder.

5 Conclusions & Future Directions

Two final datasets, as visualized in Figures 8 and 9 below have been collected, cleaned and processed from various resources and are now ready for predictive modeling.

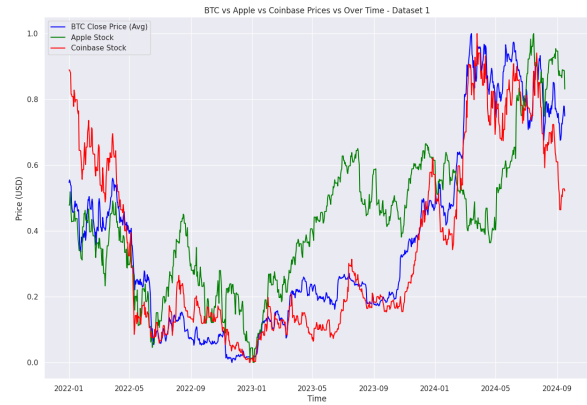


Figure 8: Dataset 1

Dataset 1 will be used to investigate long-term prediction power based on financial market features such as Apple Stock Close prices. Dataset 2 will be used to determine the short-term predictive power of social media sentiment and financial market features.

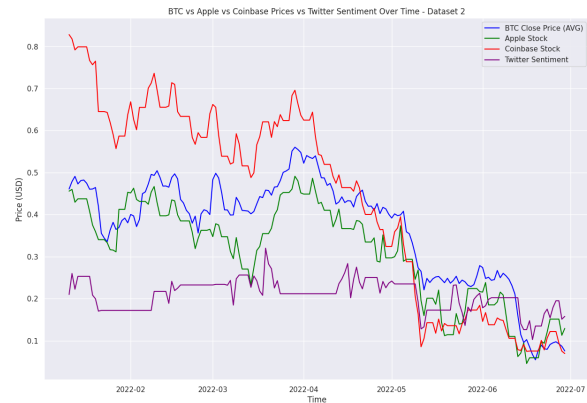


Figure 9: Dataset 2

6 References

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7 Appendix

Rejected Twitter Datasets

Tweet Source	Time Frame	Reason for Exclusion
Twitter/X API [7]	Only access to current tweets	Limited access
Kaggle: PreBit: Multimodal dataset for Bitcoin price [19]	Jan-May 2021	Timeframe too short and outdated
Kaggle: BTC Tweet Sentiment [20]	March 2023	Data available for only one day
Kaggle: Bitcoin Sentiment Analysis [21]	2021-02-05	Subset of an included dataset
Kaggle: Bitcoin Tweets 16M [22]	2016-01-01 to 2019-03-29	Too old

Table 3: Summary of Twitter sources considered and exclusion reasons

Responsible Data Reuse Checklist

For papers using a previously-published resource:

- ☒ The authors explain their choice of data, given the available resources and their known limitations (e.g. representativeness issues, biases, annotation artifacts) and any data protection issues (e.g. inclusion of sensitive health data). *See Section 2.2*
- ☒ The authors discuss whether their use of a previously-published resource is compatible with its original purpose and license, and any known limitations (e.g. if the target user group is represented in the sample). *See Section 2.3*

For papers contributing a new resource:

- ☐ The authors have the **legal basis** for processing the data and, if it is made public, for distributing it. *(Check one)*
 - ☐ The data is in public domain, and licensed for research purposes;
 - ☐ The data is used with consent of its creators or copyright holders;
 - ☐ If the data is used without consent, the paper makes the case to justify its legal basis (e.g. research performed in the public interest under GDPR). *See Section 2.4*
- ☐ The paper describes in detail the **full data collection protocol**, including collection, annotation, pre-processing, and filtering procedures. In the case that the dataset involves work by human subjects (e.g. data creation or annotation), the paper describes efforts to ensure fair compensation. *See Section 2.5*
- ☐ **Safe use** of data is ensured. *(Check all that apply)*
 - ☐ The data does not include any protected information (e.g. sexual orientation or political views under GDPR), or a specified exception applies. *See Section 2.6*
 - ☐ The paper is accompanied by a data statement describing the basic demographic and geographic characteristics of the population that is the source of the language data, and the population that it is intended to represent. *See Section 2.7*
 - ☐ If applicable: the paper describes whether any characteristics of the human subjects were self-reported (preferably) or inferred (in what way), justifying the methodology and choice of description categories. *See Section 2.8*
 - ☐ The paper discusses the harms that may ensue from the limitations of the data collection methodology, especially concerning marginalized/vulnerable populations, and specifies the scope within which the data can be used safely. *See Section 2.9*
 - ☐ If any personal data is used: the paper specifies the standards applied for its storage and processing, and any anonymization efforts. *See Section 2.10*
 - ☐ If the individual speakers remain identifiable via search: the paper discusses possible harms from misuse of this data, and their mitigation. *See Section 2.11*
- ☐ If any data or models are made public: **safe reuse** is ensured. *(Check all that apply)*
 - ☐ The data and/or pretrained models are released under a specified license that is compatible with the conditions under which access to data was granted (in particular, derivatives of data accessed for research purposes should *not* be deployed in the real world as anything other than a research prototype, especially commercially). *See Section 2.12*
 - ☐ The paper specifies the efforts to limit the potential use to circumstances in which the data/models could be used safely (such as an accompanying data/model statements). *See Section 2.13*
- ☐ The data collection protocol was **approved by the ethics review board** at the authors' institution, or such review is not applicable for specified reasons. *See Section 2.14*

Figure 10: Responsible Dataset Reuse Checklist